TU BCT — Artificial Intelligence (CT 653)

Chapter 4: Knowledge Representation — Numerical Questions & Answers

(Resolution Refutation, CNF Conversion, Forward/Backward Chaining)

Q1. Colonel West is a Criminal (Resolution Refutation)

[2082 Kartik Back Q4a | 2081 Ashwin Back Q4 | 2077 Chaitra Regular Q4 | 2075 Baisakh Back Q4]

> *The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove that Colonel West is a criminal using Resolution Refutation.*

Given Facts (in FOPL):

- American(x) ∧ Weapon(y) ∧ Hostile(z) ∧ Sells(x,y,z) → Criminal(x)
- Owns(Nono, M1) [Nono has missile M1]
- Missile(x) ∧ Owns(Nono,x) → Sells(West, x, Nono)
- Missile(x) → Weapon(x)
- Enemy(x, America) → Hostile(x)
- American(West)
- Enemy(Nono, America)
- Missile(M1)

Step 1: Convert to Clausal (CNF) Form:

```
C1: ~American(x) ∨ ~Weapon(y) ∨ ~Hostile(z) ∨ ~Sells(x,y,z) ∨ Criminal(x)
C2: Owns(Nono, M1)
C3: ~Missile(x) ∨ ~Owns(Nono,x) ∨ Sells(West, x, Nono)
C4: ~Missile(x) ∨ Weapon(x)
C5: ~Enemy(x, America) ∨ Hostile(x)
C6: American(West)
C7: Enemy(Nono, America)
C8: Missile(M1)
Negated Goal (C9): ~Criminal(West)
```

Step 2: Resolution Proof:

Step 1: Resolve C8 and C4 [x = M1]: → Weapon(M1)

Step 2: Resolve C7 and C5 [x = Nono]: → Hostile(Nono)

Step 3: Resolve C8 and C3 [x = M1]: ~Owns(Nono, M1) ∨ Sells(West, M1, Nono)

Step 4: Resolve Step 3 and C2: → Sells(West, M1, Nono)

Step 5: Resolve C1 and C6 [x = West]: ~Weapon(y) ∨ ~Hostile(z) ∨ ~Sells(West,y,z) ∨ Criminal(West)

Step 6: Resolve Step 5 and Step 1 [y = M1]: ~Hostile(z) ∨ ~Sells(West,M1,z) ∨ Criminal(West)

Step 7: Resolve Step 6 and Step 2 [z = Nono]: ~Sells(West,M1,Nono) ∨ Criminal(West)

Step 8: Resolve Step 7 and Step 4: → Criminal(West)

Step 9: Resolve Step 8 and C9 (~Criminal(West)): → [ ] (Empty Clause — Contradiction!)

Conclusion: Since the negated goal leads to contradiction, Colonel West IS a criminal. Proved!

Q2. Scrooge is Not a Child (Resolution Refutation)

[2081 Chaitra Regular Q4a | 2078 Chaitra Regular Q4]

> *Every child loves Santa. Everyone who loves Santa loves any reindeer. Rudolph is a reindeer, and Rudolph has a red nose. Anything which has a red nose is weird or is a clown. No reindeer is a clown. Scrooge does not love anything which is weird. Using Resolution Refutation, derive: Scrooge is not a child.*

Convert to FOPL and Clausal Form:

```
C1: ~Child(x) v Loves(x, Santa)                    [Every child loves Santa]
C2: ~Loves(x,Santa) v ~Reindeer(y) v Loves(x, y)   [Loves Santa → loves
reindeer]
C3: Reindeer(Rudolph)
C4: RedNose(Rudolph)
C5: ~RedNose(x) v Weird(x) v Clown(x)
C6: ~Reindeer(x) v ~Clown(x)                        [No reindeer is a clown]
C7: ~Weird(x) v ~Loves(Scrooge, x)                  [Scrooge doesn't love
weird things]
Negated Goal (C8): Child(Scrooge)
```

Resolution Proof:

Step 1: Resolve C4 and C5 [x = Rudolph]: → Weird(Rudolph) ∨ Clown(Rudolph)
Step 2: Resolve C3 and C6 [x = Rudolph]: → ~Clown(Rudolph)
Step 3: Resolve Step 1 and Step 2: → Weird(Rudolph)
Step 4: Resolve Step 3 and C7 [x = Rudolph]: → ~Loves(Scrooge, Rudolph)
Step 5: Resolve C8 and C1 [x = Scrooge]: → Loves(Scrooge, Santa)
Step 6: Resolve Step 5 and C2 [x = Scrooge]: → ~Reindeer(y) ∨ Loves(Scrooge, y)
Step 7: Resolve Step 6 and C3 [y = Rudolph]: → Loves(Scrooge, Rudolph)
Step 8: Resolve Step 7 and Step 4: → [ ] (Empty Clause — Contradiction!)

Conclusion: Scrooge is NOT a child. Proved!

Q3. Did Curiosity Kill the Cat? (Resolution Refutation)

[2080 Chaitra Regular Q4]

> *Everyone who loves all animals is loved by someone. Anyone who kills an animal is loved by no one. Jack loves all animals. Either Jack or Curiosity killed the cat. Did Curiosity kill the cat?*

Convert to FOPL and Clausal Form:

Let Tuna be the cat's name. Assumptions: Cat(Tuna), Animal(Tuna).

```
C1a: Animal(G(x)) v Loves(F(x), x)     [Skolemized from: Loves-all-animals →
loved-by-someone]
C1b: ~Loves(x, G(x)) v Loves(F(x), x) [part 2 of Skolemized clause 1]
C2:  ~Animal(y) v ~Kills(x,y) v ~Loves(z,x)   [kills animal → loved by no
one]
C3:  ~Animal(y) v Loves(Jack, y)        [Jack loves all animals]
C4:  Kills(Jack, Tuna) v Kills(Curiosity, Tuna)
C5:  Animal(Tuna)
```

```
Negated Goal (C6): ~Kills(Curiosity, Tuna)
```

Resolution Proof:

Step 1: Resolve C4 and C6: → Kills(Jack, Tuna)

Step 2: Resolve C5 and C3 [y = Tuna]: → Loves(Jack, Tuna)

Step 3: Resolve C5 and C2 [y=Tuna, x=Jack]: ~Kills(Jack,Tuna) ∨ ~Loves(z,Jack)

Step 4: Resolve Step 3 and Step 1 (Kills(Jack,Tuna)): → ~Loves(z, Jack) [Jack is loved by no one]

Step 5: From C1b, when x = Jack: ~Loves(Jack, G(Jack)) ∨ Loves(F(Jack), Jack)

Step 6: From C3, with y=G(Jack): Animal(G(Jack)) → Loves(Jack, G(Jack)) [Jack loves all animals including G(Jack)]

Step 7: From C1a with x=Jack: Animal(G(Jack)) ∨ Loves(F(Jack),Jack). If Animal(G(Jack)): from C3 → Loves(Jack,G(Jack))

Step 8: From Step 5 and Loves(Jack,G(Jack)): → Loves(F(Jack), Jack)

Step 9: Resolve Step 4 (~Loves(z,Jack)) with Step 8 (Loves(F(Jack),Jack)) [z=F(Jack)]: → [ ] (Contradiction!)

Conclusion: Our assumption ~Kills(Curiosity,Tuna) leads to contradiction. YES, Curiosity killed the cat!

Q4. Steve Likes BK301 Course (Resolution Refutation)

[2080 Ashwin Back Q5]

Steve only likes easy courses. Science courses are hard. All the courses in basket weaving department are easy. BK301 is a basket weaving course. Prove that Steve likes BK301 using Resolution Refutation.

Convert to FOPL and Clausal Form:

```
C1: ~Easy(x) ∨ Likes(Steve, x)          [Steve likes all easy courses]
C2: ~Science(x) ∨ Hard(x)                [Science courses are hard]
C3: ~BasketWeaving(x) ∨ Easy(x)          [Basket weaving courses are easy]
C4: BasketWeaving(BK301)
Negated Goal (C5): ~Likes(Steve, BK301)
```

Resolution Proof:

Step 1: Resolve C4 and C3 [x = BK301]: → Easy(BK301)

Step 2: Resolve Step 1 and C1 [x = BK301]: → Likes(Steve, BK301)

Step 3: Resolve Step 2 and C5: → [ ] (Empty Clause — Contradiction!)

Conclusion: Steve likes BK301. Proved!

Q5. Charlie is a Horse (Resolution Refutation)

[2079 Jestha Back Q4a | 2078 Poush Back Q5 | 2073 Bhadra Regular Q4 | 2071 Bhadra Q3]

Horses, cows, pigs are mammals. An offspring of a horse is a horse. Bluebeard is a horse. Bluebeard is Charlie's parent. Offspring and parent are inverse relations. Every mammal has a parent. Prove Charlie is a horse using resolution refutation.

Convert to FOPL and Clausal Form:

```
C1: ~Horse(x) ∨ Mammal(x)                [Horses are mammals]
C2: ~Cow(x) ∨ Mammal(x)                  [Cows are mammals]
C3: ~Pig(x) ∨ Mammal(x)                  [Pigs are mammals]
C4: ~Offspring(x,y) ∨ ~Horse(y) ∨ Horse(x) [Offspring of horse is horse]
C5: Horse(Bluebeard)
C6: Parent(Bluebeard, Charlie)
C7: ~Parent(x,y) ∨ Offspring(y,x)         [Inverse relation]
C8: ~Mammal(x) ∨ Parent(f(x), x)          [Every mammal has parent;
Skolemized]
Negated Goal (C9): ~Horse(Charlie)
```

Resolution Proof:

Step 1: Resolve C6 (Parent(Bluebeard,Charlie)) and C7 [x=Bluebeard, y=Charlie]: → Offspring(Charlie, Bluebeard)

Step 2: Resolve Step 1 and C4 [x=Charlie, y=Bluebeard]: ~Horse(Bluebeard) ∨ Horse(Charlie)

Step 3: Resolve Step 2 and C5 (Horse(Bluebeard)): → Horse(Charlie)

Step 4: Resolve Step 3 and C9 (~Horse(Charlie)): → [ ] (Empty Clause — Contradiction!)

Conclusion: Charlie is a horse. Proved!

Q6. Taarey is a Cow (Resolution Refutation)

[2078 Baishakh Back Q4]

> *Cow, Buffalo, Bull are mammals. An offspring of a Cow is a Cow. Kali is a Cow. Kali is Taarey's parent. Offspring and parent are inverse relations. Every mammal has a parent. Prove that Taarey is a Cow using resolution refutation.*

Convert to Clausal Form:

```
C1: ~Cow(x) ∨ Mammal(x)
C2: ~Offspring(x,y) ∨ ~Cow(y) ∨ Cow(x)    [Offspring of cow is cow]
C3: Cow(Kali)
C4: Parent(Kali, Taarey)
C5: ~Parent(x,y) ∨ Offspring(y,x)          [Inverse relation]
C6: ~Mammal(x) ∨ Parent(f(x), x)           [Skolemized]
Negated Goal (C7): ~Cow(Taarey)
```

Resolution Proof:

Step 1: Resolve C4 and C5 [x=Kali, y=Taarey]: → Offspring(Taarey, Kali)

Step 2: Resolve Step 1 and C2 [x=Taarey, y=Kali]: → ~Cow(Kali) ∨ Cow(Taarey)

Step 3: Resolve Step 2 and C3 (Cow(Kali)): → Cow(Taarey)

Step 4: Resolve Step 3 and C7: → [ ] (Contradiction!)

Conclusion: Taarey is a Cow. Proved!

Q7. John is a Light Sleeper — No Mice (Resolution Refutation)

[2076 Baisakh Back Q4]

> *All hounds howl at night. Anyone who has any cats will not have any mice. Light sleepers do not have anything which howls at night. John has either a cat or a hound. Prove: 'If John is a light sleeper, then John does not have any mice.'*

Convert to FOPL and Clausal Form:

```
C1: ~Hound(x) ∨ Howls(x)
C2: ~Has(x,y) ∨ ~Cat(y) ∨ ~Has(x,z) ∨ ~Mouse(z)  [Has cat → no mice]
C3: ~LightSleeper(x) ∨ ~Howls(y) ∨ ~Has(x,y)      [Light sleepers have
nothing that howls]
C4a: Cat(C) ∨ Hound(C)                              [Skolem constant C for
John's pet]
C4b: Has(John, C)
Negated Conclusion:
  C5: LightSleeper(John)
  C6: Has(John, M)   [M is a Skolem constant for some mouse]
  C7: Mouse(M)
```

Resolution Proof (by case analysis):

Case 1: C is a Cat (Cat(C)):

Step 1: C4b (Has(John,C)), Cat(C), C6 (Has(John,M)), C7 (Mouse(M)) — apply C2 [x=John, y=C, z=M]

C2 becomes: ~Has(John,C) ∨ ~Cat(C) ∨ ~Has(John,M) ∨ ~Mouse(M)

All four negatives are contradicted by their positives → [ ] Contradiction!

Case 2: C is a Hound (Hound(C)):

Step 1: Resolve Hound(C) and C1 [x=C]: → Howls(C)

Step 2: Resolve C5 (LightSleeper(John)), Howls(C) and C3 [x=John, y=C]: → ~Has(John, C)

Step 3: Resolve Step 2 and C4b (Has(John,C)): → [ ] Contradiction!

Both cases lead to contradiction → If John is a light sleeper, he has no mice. Proved!

Q8. John Likes Peanuts (Resolution Refutation)

[2074 Bhadra Regular Q4a | 2072 Ashwin Regular Q5 | 2075 Baisakh Back Q4 (Donald version)]

> *John likes all kinds of food. Apples are food. Chicken is food. Anything anyone eats and isn't killed by is food. Bill eats peanuts and is still alive. Sue eats everything Bill eats. Prove that John likes peanuts.*

Convert to FOPL and Clausal Form:

```
C1:  ~Food(x) ∨ Likes(John, x)
C2:  Food(Apple)
C3:  Food(Chicken)
```

```
C4:  ~Eats(x,y) ∨ Killed(x) ∨ Food(y)    [if eaten and not killed, then food]
C5:  Eats(Bill, Peanut)
C6:  ~Killed(Bill)                        [Bill is still alive]
C7:  ~Eats(Bill, x) ∨ Eats(Sue, x)
Negated Goal (C8): ~Likes(John, Peanut)
```

Resolution Proof:

Step 1: Resolve C5 and C4 [x=Bill, y=Peanut]: → Killed(Bill) ∨ Food(Peanut)
Step 2: Resolve Step 1 and C6 (~Killed(Bill)): → Food(Peanut)
Step 3: Resolve Step 2 and C1 [x=Peanut]: → Likes(John, Peanut)
Step 4: Resolve Step 3 and C8: → [ ] (Empty Clause — Contradiction!)

Conclusion: John likes peanuts. Proved!

Q9. Did Marcus Hate Caesar? (Resolution Refutation)

[2072 Magh New Back Q5]

> *All pompeians are Romans. All Romans were either loyal to Caesar or hated him. Everyone is loyal to someone. People only try to assassinate rulers they are not loyal to. Marcus tried to assassinate Caesar. Marcus was a pompeian. Find: Did Marcus hate Caesar?*

Convert to FOPL and Clausal Form:

```
C1: ~Pompeian(x) ∨ Roman(x)
C2: ~Roman(x) ∨ LoyalTo(x, Caesar) ∨ Hates(x, Caesar)
C3: LoyalTo(x, f(x))                          [Skolemized: everyone loyal
to someone]
C4: ~TryAssassinate(x,y) ∨ ~Ruler(y) ∨ ~LoyalTo(x,y)
C5: TryAssassinate(Marcus, Caesar)
C6: Pompeian(Marcus)
C7: Ruler(Caesar)
Negated Goal (C8): ~Hates(Marcus, Caesar)
```

Resolution Proof:

Step 1: Resolve C6 and C1 [x=Marcus]: → Roman(Marcus)
Step 2: Resolve Step 1 and C2 [x=Marcus]: → LoyalTo(Marcus, Caesar) ∨ Hates(Marcus, Caesar)
Step 3: Resolve Step 2 and C8 (~Hates(Marcus,Caesar)): → LoyalTo(Marcus, Caesar)
Step 4: Resolve C5 and C4 [x=Marcus, y=Caesar]: → ~Ruler(Caesar) ∨ ~LoyalTo(Marcus, Caesar)
Step 5: Resolve Step 4 and C7 (Ruler(Caesar)): → ~LoyalTo(Marcus, Caesar)
Step 6: Resolve Step 3 and Step 5: → [ ] (Contradiction!)

Conclusion: Marcus DID hate Caesar. Proved!

Q10. Ram is Naughty (Resolution Refutation)

[2070 Bhadra Regular Q4]

> *All oversmart persons are stupid. Children of oversmart persons are naughty. Ram is a child of Hari. Hari is oversmart. Show that Ram is naughty using FOPL based Resolution Refutation method.*

Convert to FOPL and Clausal Form:

```
C1: ~Oversmart(x) ∨ Stupid(x)
C2: ~Oversmart(x) ∨ ~Child(y, x) ∨ Naughty(y)   [Children of oversmart are naughty]
C3: Child(Ram, Hari)                             [Ram is child of Hari]
C4: Oversmart(Hari)
Negated Goal (C5): ~Naughty(Ram)
```

Resolution Proof:

Step 1: Resolve C4 and C2 [x=Hari]: → ~Child(y, Hari) ∨ Naughty(y)

Step 2: Resolve Step 1 and C3 [y=Ram]: → Naughty(Ram)

Step 3: Resolve Step 2 and C5: → [ ] (Contradiction!)

Conclusion: Ram is naughty. Proved!

Q11. If John Does Not Study, Mary Does Not Love John (Resolution Refutation)

[2073 Magh New Back Q4]

> *Given: (i) Anyone whom Mary loves is a football star. (ii) Any student who does not pass does not play. (iii) John is a student. (iv) Any student who does not study does not pass. (v) Anyone who does not play is not a football star. Prove: If John does not study, then Mary does not love John.*

Convert to FOPL and Clausal Form:

```
C1: ~Loves(Mary, x) ∨ FootballStar(x)
C2: ~Student(x) ∨ Pass(x) ∨ ~Play(x)        [does not pass → does not play]
C3: Student(John)
C4: ~Student(x) ∨ Study(x) ∨ ~Pass(x)       [does not study → does not pass]
C5: Play(x) ∨ ~FootballStar(x)               [does not play → not football star]
Negated Conclusion:
  C6: ~Study(John)
  C7: Loves(Mary, John)
```

Resolution Proof:

Step 1: Resolve C7 and C1 [x=John]: → FootballStar(John)

Step 2: Resolve Step 1 and C5 [x=John]: → Play(John)

Step 3: Resolve C3 and C4 [x=John]: → Study(John) ∨ ~Pass(John)

Step 4: Resolve Step 3 and C6 (~Study(John)): → ~Pass(John)
Step 5: Resolve C3 and C2 [x=John]: → Pass(John) ∨ ~Play(John)
Step 6: Resolve Step 5 and Step 4 (~Pass(John)): → ~Play(John)
Step 7: Resolve Step 6 and Step 2 (Play(John)): → [ ] (Contradiction!)

Conclusion: If John does not study, then Mary does not love John. Proved!

Q12. Block World — Supports(Book, Cup) (Resolution Refutation)

[2071 Bhadra Regular/Back Q3]

> *If X is on the top of Y, Y supports X. If X is above Y and they are touching each other, X is on the top of Y. A cup is above a book. A cup is touching a book. Show that Supports(Book, Cup) is true.*

Convert to Clausal Form:

```
C1: ~OnTop(x,y) ∨ Supports(y,x)
C2: ~Above(x,y) ∨ ~Touching(x,y) ∨ OnTop(x,y)
C3: Above(Cup, Book)
C4: Touching(Cup, Book)
Negated Goal (C5): ~Supports(Book, Cup)
```

Resolution Proof:

Step 1: Resolve C3 and C2 [x=Cup, y=Book]: → ~Touching(Cup,Book) ∨ OnTop(Cup,Book)
Step 2: Resolve Step 1 and C4: → OnTop(Cup, Book)
Step 3: Resolve Step 2 and C1 [x=Cup, y=Book]: → Supports(Book, Cup)
Step 4: Resolve Step 3 and C5: → [ ] (Contradiction!)

Conclusion: Supports(Book, Cup) is true. Proved!

Q13. CNF Conversion: Everyone Who Loves All Animals Is Loved By Someone

[2080 Chaitra Regular Q4 | 2079 Chaitra Regular Q5 | 2075 Bhadra Regular Q4]

> *Convert the following sentence to Conjunctive Normal Form (CNF): 'Everyone who loves all animals is loved by someone.' Also represent in FOPL.*

FOPL Representation:

```
∀x [ (∀y Animal(y) → Loves(x,y)) → ∃z Loves(z,x) ]
```

Steps to Convert to CNF:

Step 1: Eliminate Implication (A→B becomes ¬A∨B)

```
∀x [ ¬(∀y Animal(y) → Loves(x,y)) ∨ ∃z Loves(z,x) ]
```

```
= ∀x [ ¬(∀y ¬Animal(y) ∨ Loves(x,y)) ∨ ∃z Loves(z,x) ]
```

Step 2: Move Negation Inward (De Morgan's Law)

```
∀x [ (∃y Animal(y) ∧ ¬Loves(x,y)) ∨ ∃z Loves(z,x) ]
```

Step 3: Rename Variables (Standardize Apart)

```
∀x [ (∃y Animal(y) ∧ ¬Loves(x,y)) ∨ ∃z Loves(z,x) ]
```

Step 4: Skolemize (Replace Existentials with Skolem Functions)

Since y and z depend on x (under ∀x), they become Skolem functions G(x) and F(x):

```
∀x [ (Animal(G(x)) ∧ ¬Loves(x, G(x))) ∨ Loves(F(x), x) ]
```

Step 5: Drop Universal Quantifier

```
(Animal(G(x)) ∧ ¬Loves(x, G(x))) ∨ Loves(F(x), x)
```

Step 6: Distribute (∨ over ∧) to get CNF

```
= (Animal(G(x)) ∨ Loves(F(x),x)) ∧ (¬Loves(x,G(x)) ∨ Loves(F(x),x))
```

CNF Clauses: Clause 1: Animal(G(x)) ∨ Loves(F(x), x) Clause 2: ¬Loves(x, G(x)) ∨ Loves(F(x), x)

Q14. CNF Conversion of Two Logical Formulas

[2079 Ashwin Back Q4b]

> *Transform each of the following sentences into CNF: (i) ~(P & Q) V (P V S) → R (ii) P → ((Q & ~R) ↔ S)*

(i) ~(P ∧ Q) ∨ (P ∨ S) → R

Step 1: Eliminate Implication (A → B becomes ¬A ∨ B)

```
¬[~(P ∧ Q) ∨ (P ∨ S)] ∨ R
```

Step 2: Move Negation Inward

```
= [~~(P ∧ Q) ∧ ~(P ∨ S)] ∨ R
= [(P ∧ Q) ∧ (~P ∧ ~S)] ∨ R
= [P ∧ Q ∧ ~P ∧ ~S] ∨ R
```

Step 3: Simplify

Since P ∧ ¬P = False, the expression [P ∧ Q ∧ ¬P ∧ ¬S] = False

```
= False ∨ R = R
```

CNF Result (i): { R } (The formula simplifies to just R)

(ii) P → ((Q ∧ ~R) ↔ S)

Step 1: Expand Biconditional (A ↔ B becomes (A→B) ∧ (B→A))

```
P → [((Q ∧ ¬R) → S) ∧ (S → (Q ∧ ¬R))]
```

Step 2: Eliminate Implications

```
P → [(¬(Q ∧ ¬R) ∨ S) ∧ (¬S ∨ (Q ∧ ¬R))]
= P → [(¬Q ∨ R ∨ S) ∧ (¬S ∨ Q) ∧ (¬S ∨ ¬R)]
```

Step 3: Eliminate outer implication

```
¬P ∨ [(¬Q ∨ R ∨ S) ∧ (¬S ∨ Q) ∧ (¬S ∨ ¬R)]
```

Step 4: Distribute ∨ over ∧

```
(¬P ∨ ¬Q ∨ R ∨ S) ∧ (¬P ∨ ¬S ∨ Q) ∧ (¬P ∨ ¬S ∨ ¬R)
```

CNF Clauses (ii): Clause 1: ¬P ∨ ¬Q ∨ R ∨ S Clause 2: ¬P ∨ ¬S ∨ Q Clause 3: ¬P ∨ ¬S ∨ ¬R

Q15. Forward Chaining — Colonel West is a Criminal

[2081 Ashwin Back Q4]

> *Using the Colonel West facts, prove that Colonel West is a criminal using Forward Chaining.*

Knowledge Base (Rules & Facts):

Rules:

```
R1: American(x) ∧ Weapon(y) ∧ Hostile(z) ∧ Sells(x,y,z) → Criminal(x)
R2: Missile(x) ∧ Owns(Nono,x) → Sells(West, x, Nono)
R3: Missile(x) → Weapon(x)
R4: Enemy(x, America) → Hostile(x)
```

Facts:

```
F1: American(West)
F2: Enemy(Nono, America)
F3: Missile(M1)
F4: Owns(Nono, M1)
```

Forward Chaining Steps:

Start from facts and fire applicable rules until goal is derived:

Step 1: Apply R4 with F2 (Enemy(Nono,America)): → Hostile(Nono) [NEW FACT]

Step 2: Apply R3 with F3 (Missile(M1)): → Weapon(M1) [NEW FACT]

Step 3: Apply R2 with F3 (Missile(M1)) and F4 (Owns(Nono,M1)): → Sells(West, M1, Nono) [NEW FACT]

Step 4: Apply R1 with F1 (American(West)) ∧ Weapon(M1) ∧ Hostile(Nono) ∧ Sells(West,M1,Nono): → Criminal(West) [GOAL REACHED!]

Conclusion: Criminal(West) derived by Forward Chaining. Colonel West is a criminal!

Q16. Everyone Who Loves All Animals Prove: Curiosity & Jack (CNF + Resolution)

[2080 Chaitra Regular Q4 — Part: Convert 'everyone loves all animals' to CNF for resolution]

> *Convert the given sentences to FOPL, list down the steps for converting to CNF, and answer 'Did Curiosity kill the cat?' by Resolution Refutation. Sentences: (a) Everyone who loves all animals is loved by someone. (b) Anyone who kills an animal is loved by no one. (c) Jack loves all animals. (d) Either Jack or Curiosity killed the cat.*

Step A: Convert All Sentences to FOPL

```
(a) ∀x [(∀y Animal(y) → Loves(x,y)) → ∃z Loves(z,x)]
(b) ∀x ∀y [Animal(y) ∧ Kills(x,y) → ∀z ¬Loves(z,x)]
(c) ∀y Animal(y) → Loves(Jack, y)
(d) Kills(Jack, Tuna) ∨ Kills(Curiosity, Tuna)   [Tuna = the cat]
    Animal(Tuna)
```

Step B: Convert to CNF

```
From (a) [Skolemize: G(x), F(x)]:
  A1: Animal(G(x)) ∨ Loves(F(x), x)
  A2: ¬Loves(x, G(x)) ∨ Loves(F(x), x)
From (b):
  B1: ¬Animal(y) ∨ ¬Kills(x,y) ∨ ¬Loves(z,x)
From (c):
  C1: ¬Animal(y) ∨ Loves(Jack, y)
(d): D1: Kills(Jack,Tuna) ∨ Kills(Curiosity,Tuna)
     D2: Animal(Tuna)
```

Step C: Resolution Proof (Goal: Kills(Curiosity,Tuna))

Negate goal: ¬Kills(Curiosity, Tuna)

Step 1: Resolve ¬Kills(Curiosity,Tuna) and D1: → Kills(Jack,Tuna)

Step 2: Resolve D2 and C1 [y=Tuna]: → Loves(Jack,Tuna)

Step 3: Resolve D2 and B1 [y=Tuna, x=Jack]: → ¬Kills(Jack,Tuna) ∨ ¬Loves(z,Jack)

Step 4: Resolve Step 3 and Step 1 (Kills(Jack,Tuna)): → ¬Loves(z, Jack) [Jack loved by no one]

Step 5: From A1 [x=Jack]: Animal(G(Jack)) ∨ Loves(F(Jack),Jack)

Step 6: If Animal(G(Jack)), then from C1: Loves(Jack, G(Jack)); use A2[x=Jack]: ¬Loves(Jack,G(Jack)) ∨ Loves(F(Jack),Jack) → Loves(F(Jack),Jack)

Step 7: Resolve Step 4 [z=F(Jack)] and Step 6: → [ ] Contradiction!

Conclusion: Curiosity killed the cat. Proved!

Q17. John Likes Peanuts — Also Prove Sue Eats Peanuts

[2072 Ashwin Regular Q5 extended / 2074 Bhadra Q4a extended]

> *Prove John likes peanuts AND that Sue eats peanuts using the food resolution KB.*

(a) Prove: Likes(John, Peanut) — see Q8 above

Already proved above. Key steps: Eats(Bill,Peanut) ∧ ¬Killed(Bill) → Food(Peanut) → Likes(John,Peanut). □

(b) Prove: Eats(Sue, Peanut)

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C5: Eats(Bill, Peanut)
C7: ¬Eats(Bill, x) ∨ Eats(Sue, x)
Negated Goal: ¬Eats(Sue, Peanut)
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Step 1: Resolve C5 (Eats(Bill,Peanut)) and C7 [x=Peanut]: → Eats(Sue, Peanut)

Step 2: Resolve Step 1 and negated goal: → [ ] Contradiction!

Conclusion: Sue eats peanuts. Proved!

Summary of All Numerical Questions

Q#	Question Topic	Appeared In	Method
1	Colonel West is a Criminal	2082K, 2081A, 2077C, 2075B	Resolution Refutation
2	Scrooge is not a Child	2081C, 2078C	Resolution Refutation
3	Did Curiosity kill the cat?	2080C, 2080A	Resolution Refutation
4	Steve likes BK301	2080A	Resolution Refutation
5	Charlie is a Horse	2079J, 2078P, 2073B, 2071B	Resolution Refutation
6	Taarey is a Cow	2078B	Resolution Refutation
7	John (light sleeper) no mice	2076B	Resolution Refutation
8	John likes Peanuts	2074B, 2072A	Resolution Refutation
9	Marcus hated Caesar	2072M	Resolution Refutation
10	Ram is Naughty	2070B	Resolution Refutation
11	¬Study(John)→¬Loves(Mary,John)	2073M	Resolution Refutation
12	Supports(Book, Cup)	2071B	Resolution Refutation
13	CNF: Loves-all-animals	2080C, 2079C, 2075B	CNF Conversion
14	CNF: Two formulas	2079A	CNF Conversion
15	Forward Chaining: Col. West	2081A	Forward Chaining
16	Full Curiosity kills cat (CNF+Res)	2080C	CNF + Resolution
17	John/Sue and Peanuts	2074B, 2072A	Resolution Refutation